U.S. FOOD & DRUG ADMINISTRATION

AI/ML in Drug Development and Evaluation: Past, Present, and Paths Forward

Hana Lee, Ph.D.

hana.lee@fda.hhs.gov

Office of Biostatistics
Center for Drug Evaluation and Research
US Food and Drug Administration

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Disclaimer

This presentation reflects the view of the author and should not be construed to represent FDA's views or policies.

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Patrizia Cavazzoni

Director,

CDER, FDA

Tala Fakhouri

Associate Director,

Data Science and Artificial Intelligence Policy,

Office of Medical Policy,

CDER, FDA

Outline

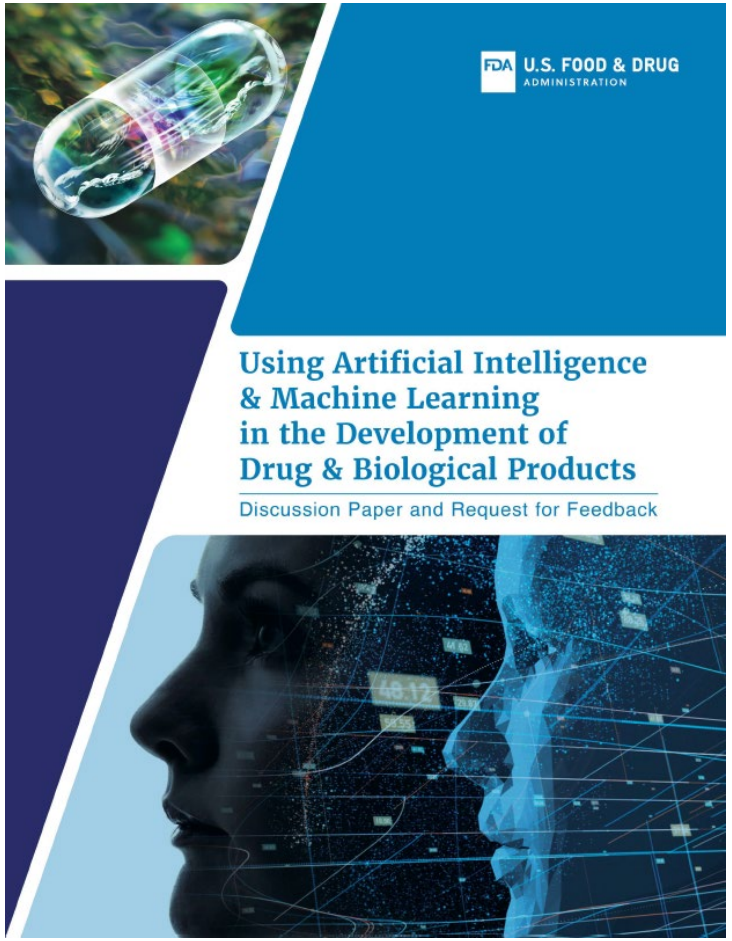
1. Definition of AI and ML
2. Overview
   - AI/ML across the drug development process
   - Regulatory programs
   - Landscape assessment and regulatory perspective
3. Challenges and efforts

Part 1

Definitions

Definitions: AI and ML

FDA CDER, CBER and CDRH issued a **discussion paper** to explore relevant considerations for the use of AI/ML in the development of drugs and biological products.



Artificial Intelligence (AI):

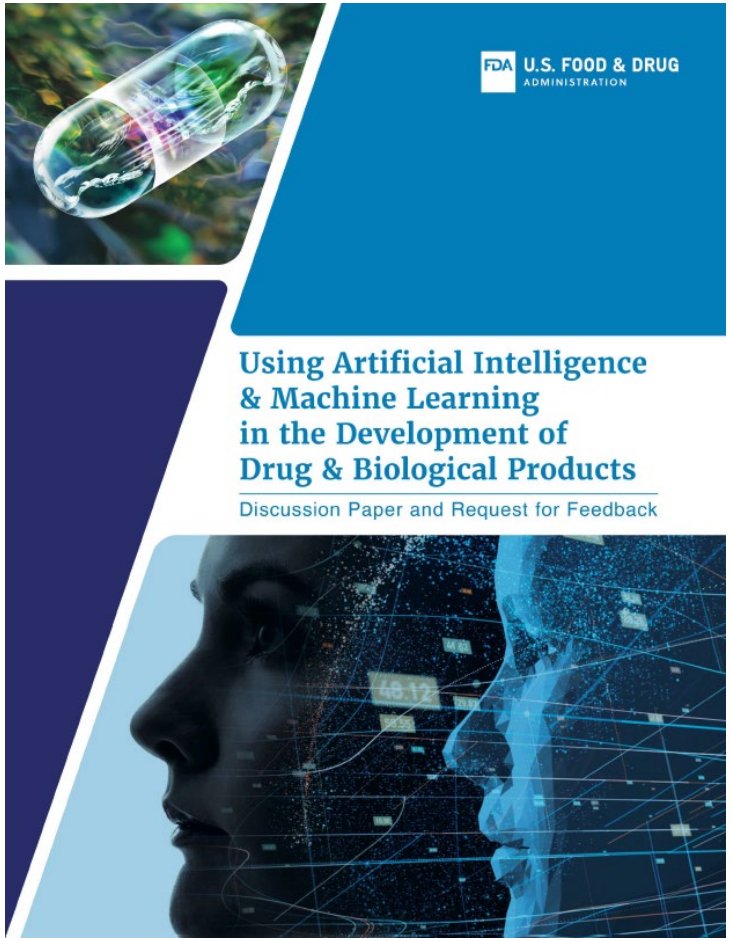
AI is a machine-based system that can, for a given set of human-defined objectives, make predictions, recommendations, or decisions influencing real or virtual environments.

https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-software-medical-device

Definitions: AI and ML

Machine Learning (ML):

A set of techniques that can be used to train AI algorithms to improve performance of a task based on data.

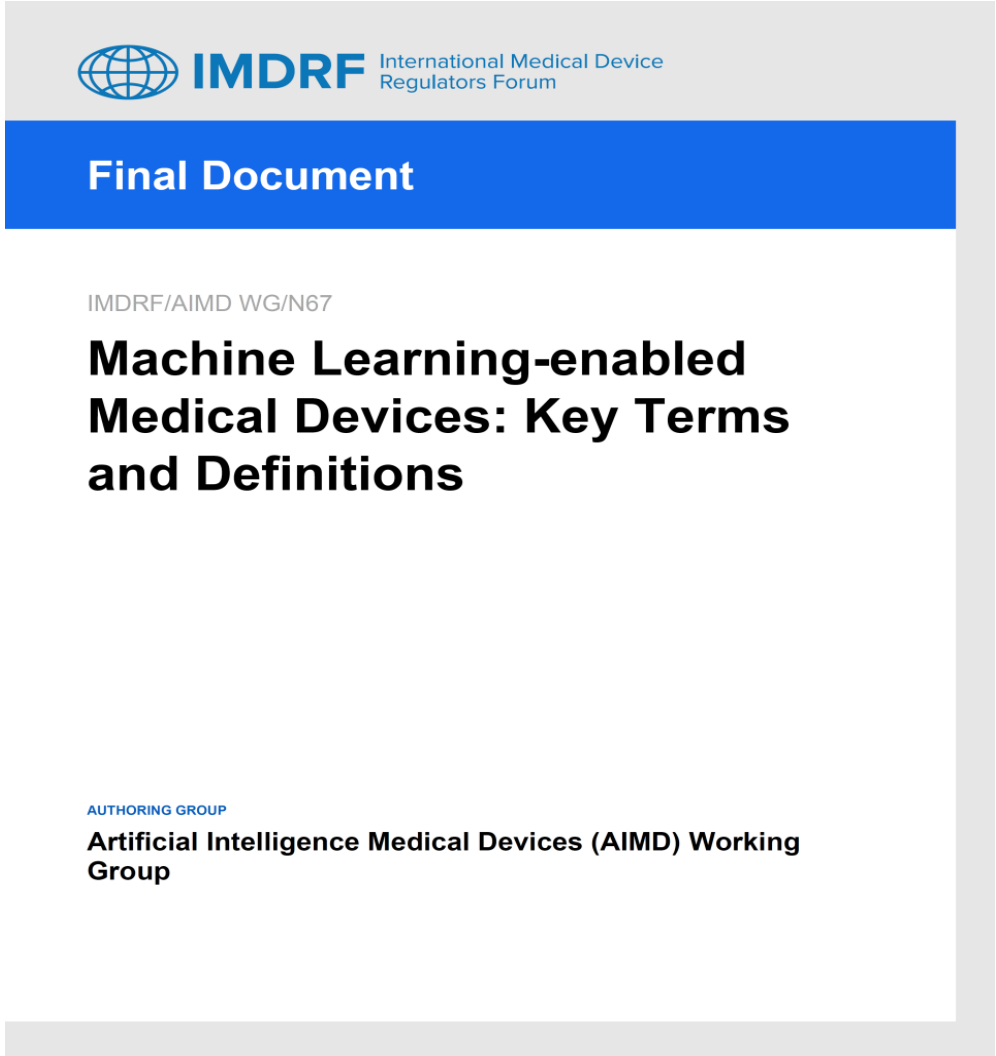


https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-software-medical-device

Definitions: AI and ML

Artificial Intelligence (AI):

A branch of computer science, statistics, and engineering that uses algorithms or models to perform tasks and exhibit behaviors such as learning, making decisions and making predictions


IMDRF International Medical Device Regulators Forum

Final Document

IMDRF/AIMD WG/N67

Machine Learning-enabled Medical Devices: Key Terms and Definitions

AUTHORING GROUP

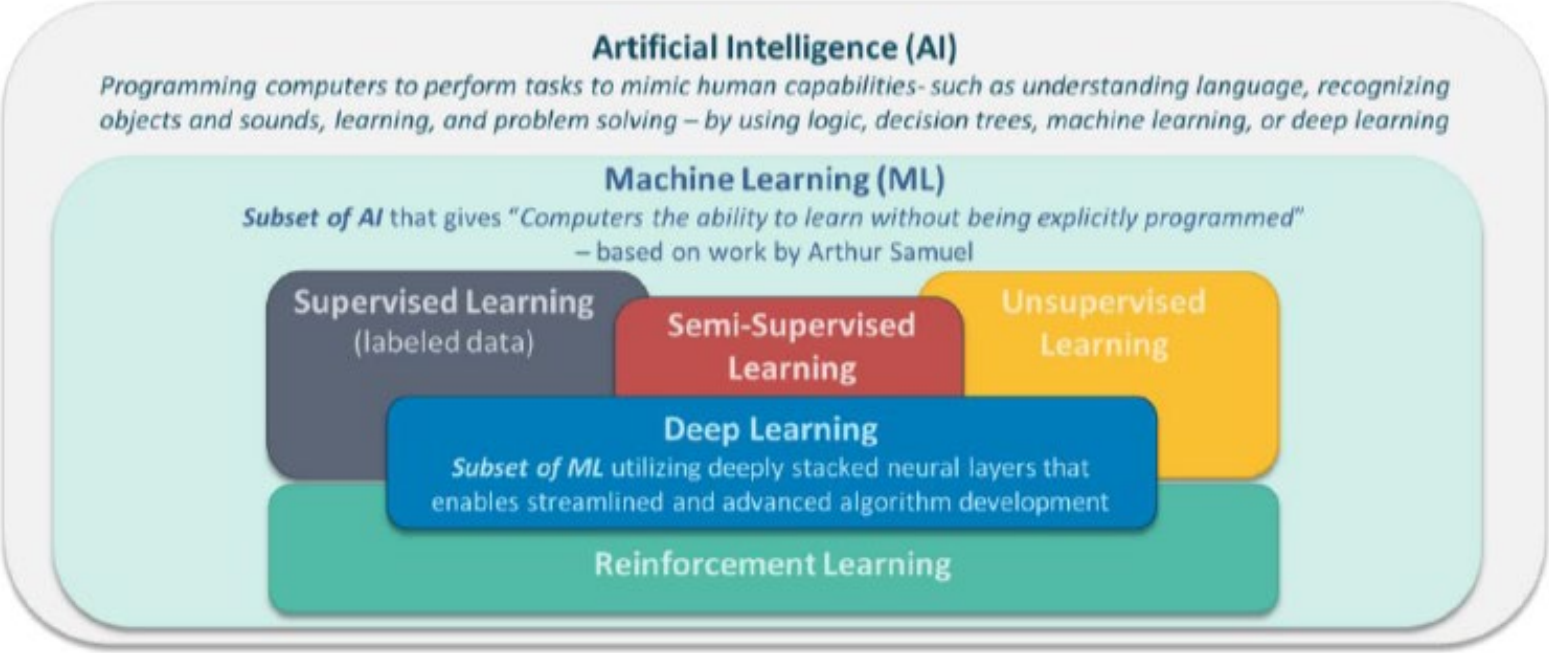
Artificial Intelligence Medical Devices (AIMD) Working Group

https://www.imdrf.org/documents/machine-learning-enabled-medical-devices-key-terms-and-definitions

Definitions: AI and ML

Machine Learning (ML):

A subset of AI that allows ML models to be developed by ML training algorithms through analysis of data, without models being explicitly programmed



※ The descriptions within the diagram are not definitions, and are included to convey a general sense of the technology.

https://www.imdrf.org/documents/machine-learning-enabled-medical-devices-key-terms-and-definitions

Part 2

Overview

AI Across the Drug Development Process

FDA

Discovery

- Drug Target Identification, Selection, and Prioritization
- Compound Screening and Design

Nonclinical Research

- PK/PD and toxicologic studies
- Dose range finding

Clinical Research

Image source: cbinsights.com

- Dose range finding
- Site selection
- Recruitment and Retention
- Adherence
- Data collection, management, and analysis
- RWD analyses
- Clinical endpoint assessment

Manufacturing and Postmarket Safety Monitoring

- Advanced pharmaceutical manufacturing
- Post-market safety surveillance or pharmacovigilance (PV)

Source: Patrizia Cavazzoni, Keynote address from Hybrid Public Workshop on AI in Drug & Biological Product Development

Regulatory Programs

CDER has several pathways for engagement that depend on AI's context of use

Examples of Engagement Pathways

Nonclinical Research

Clinical Research

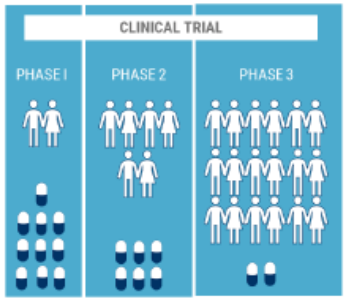


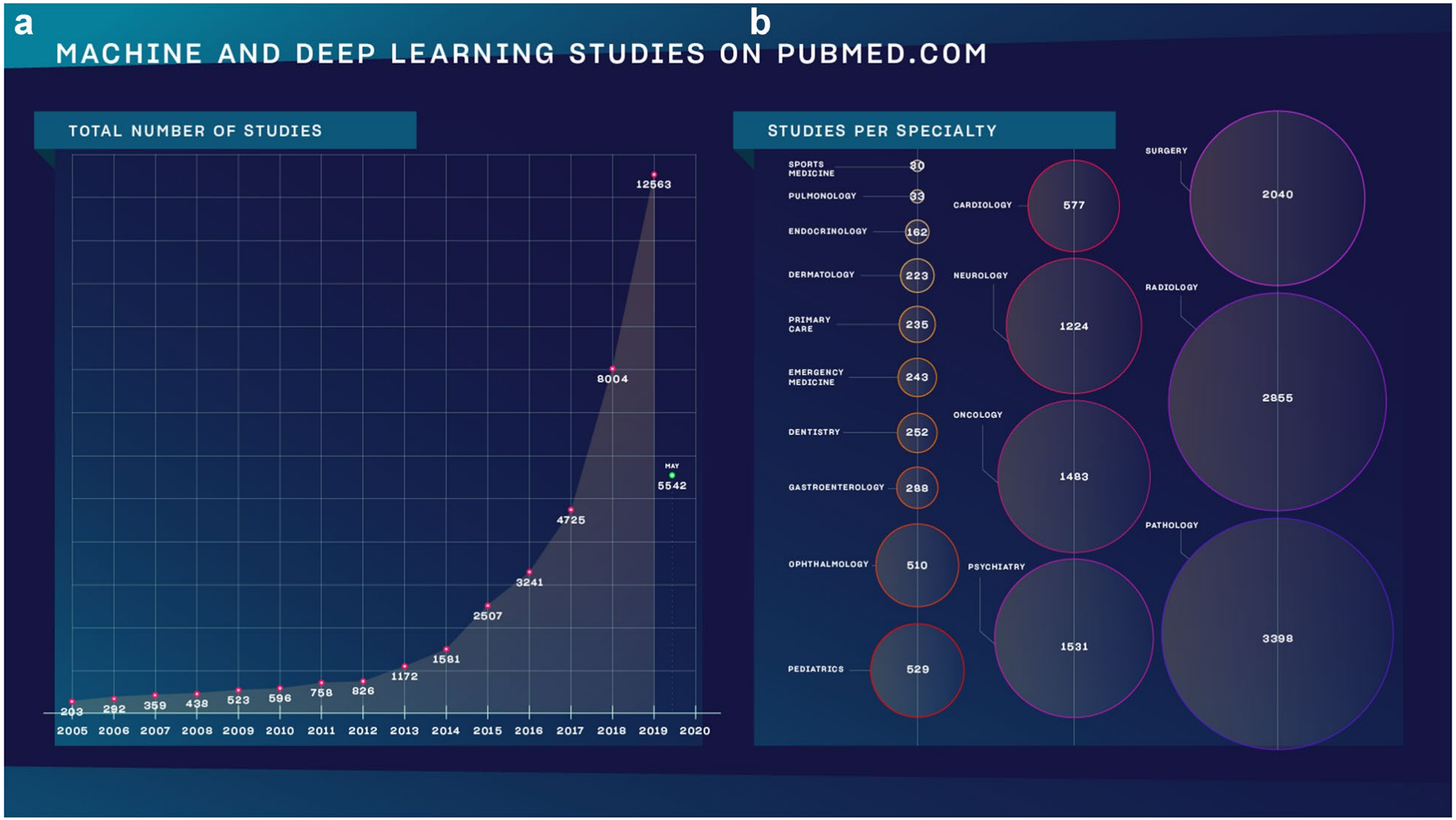
Image source: cbinsights.com

Manufacturing and Postmarket Safety Monitoring

- Model-Informed Drug Development Program
- Real-World Evidence (RWE) Program
- Digital Health Technology (DHT) Program
- Complex Innovative Trial Design Meeting Program
- CDER Center for Clinical Trial Innovation (C3TI)
- Emerging Drug Safety Technology Program (EDSTP)
- CDER's Emerging Technology Program (ETP)

Source: Patrizia Cavazzoni, Keynote address from Hybrid Public Workshop on AI in Drug & Biological Product Development

Number of medical AI studies by medical specialties (2010 – 2020)



Meskó, B., & Görög, M. (2020). A short guide for medical professionals in the era of artificial intelligence. NPJ digital medicine, 3(1), 126.

Landscape Analysis: Regulatory Submissions with AI/ML Components

Count of regulatory submissions for drug development with key terms "machine learning" or "artificial intelligence" from 2016 to 2021

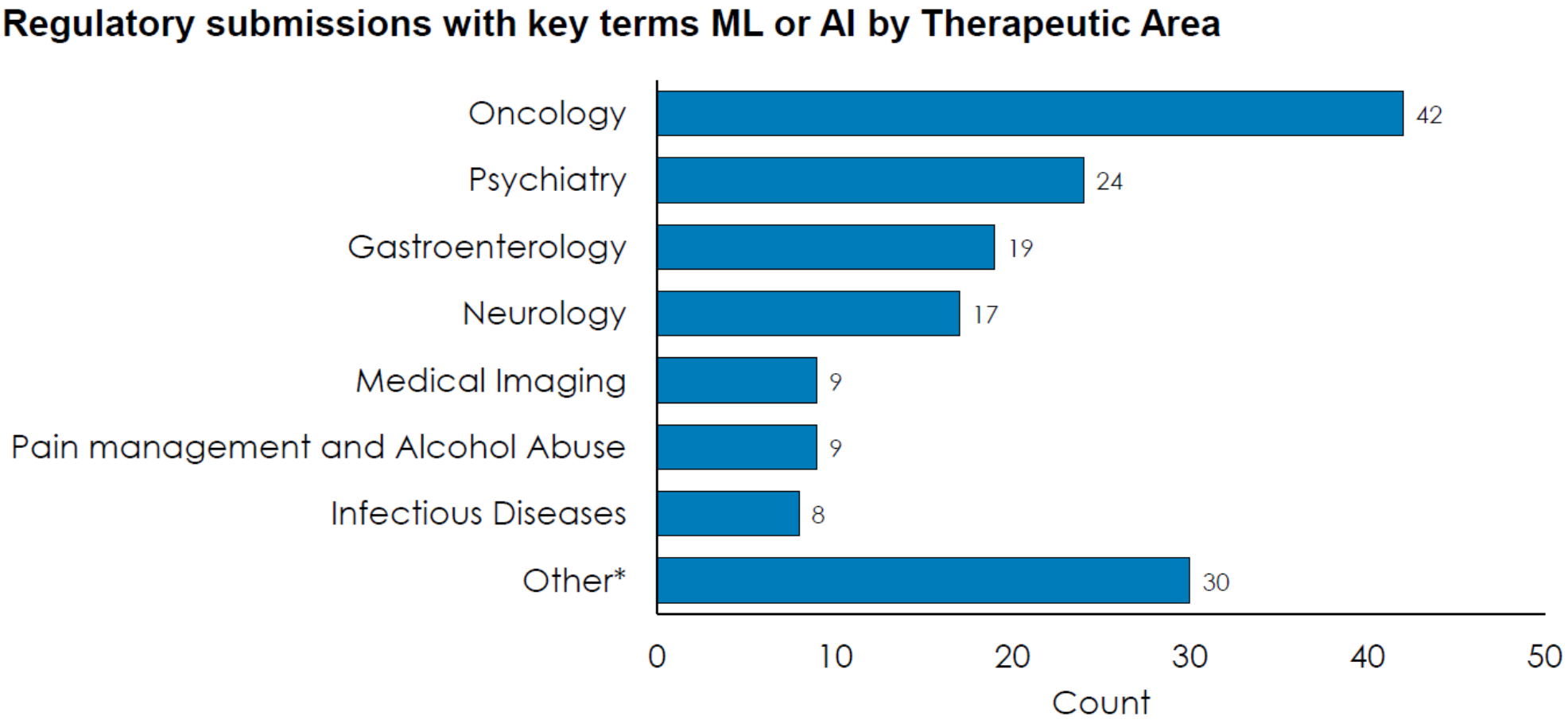
	Year					
Submission Type (n)	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**
IND	1	1	2	5	11	128
NDA, ANDA, BLA	-	-	1	2	2	2
DDT, CPIM	-	-	-	-	1	2
	Year					
Drug Development Stage (n)	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**
Discovery and Development	-	-	-	-	1	3
Preclinical Research	-	-	-	-	-	8
Clinical Research	1	1	3	5	12	118
Post-Market Safety Monitoring	-	-	-	2	1	3

ABBREVIATIONS: Investigational New Drug (IND); New Drug Application (NDA), Abbreviated New Drug Application (ANDA), Biologics License Application (BLA); Drug Development Tool (DDT) Qualification Programs, Critical Path Innovation Meeting (CPIM)

SOURCE: Internal databases maintained by the FDA Center for Drug Evaluation and Research (CDER)

Liu et al. (2023). Landscape analysis of the application of artificial intelligence and machine learning in regulatory submissions for drug development from 2016 to 2021. Clinical pharmacology and therapeutics, 113(4), 771-774.

Landscape Analysis: Regulatory Submissions with AI/ML Components



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AI/ML in Regulatory Review and Research

FDA reviewers/researchers have been applying AI/ML algorithms for various applications:

- Classification and regression trees for determining the optimal dosage range
- Predicting drug toxicity
- Predict clinical sites that will qualify for FDA inspection
- Hierarchical clustering of pharmacokinetic (PK) curves
- ML for regulatory drug safety evaluation including the use of NLP to analyze safety reports received by the FDA Adverse Event Reporting System (FAERS)

AI Across the Drug Development Process

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Source: Patrizia Cavazzoni, Keynote address from Hybrid Public Workshop on AI in Drug & Biological Product Development

Sentinel

FDA-Sentinel: Data & Scientific Partners

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Mass General Brigham

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Duke University School of Medicine

VANDERBILT UNIVERSITY MEDICAL CENTER
Department of Biomedical Informatics

DEPARTMENT OF POPULATION MEDICINE
Harvard Pilgrim Health Care Institute

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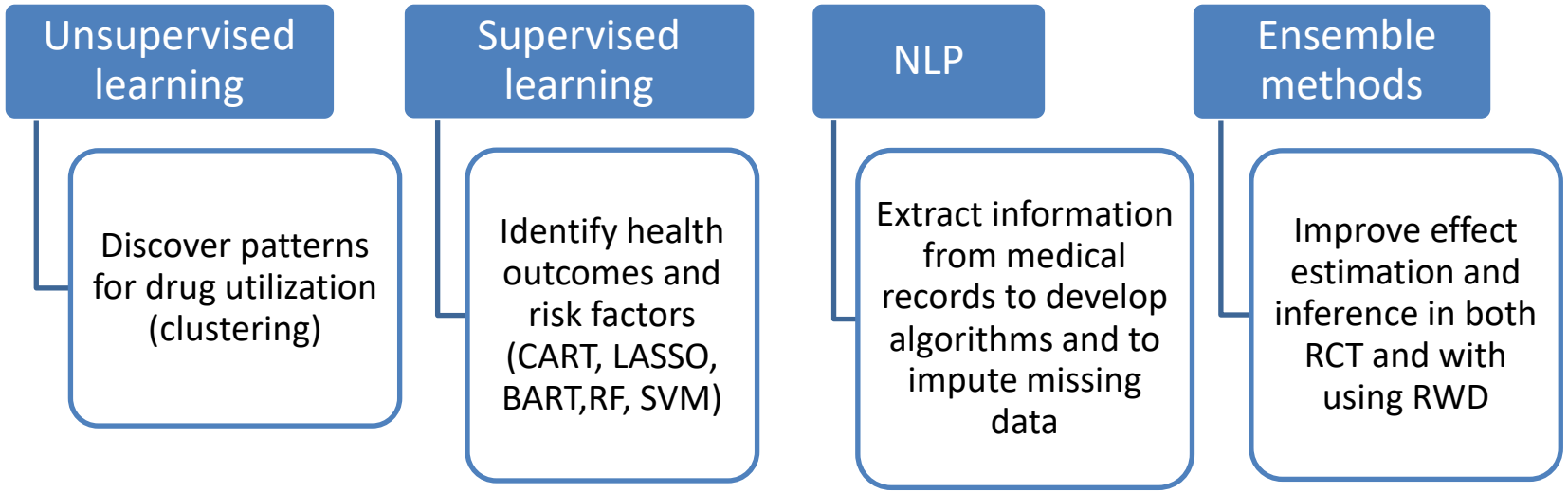
ML in Regulatory Drug Safety Evaluation

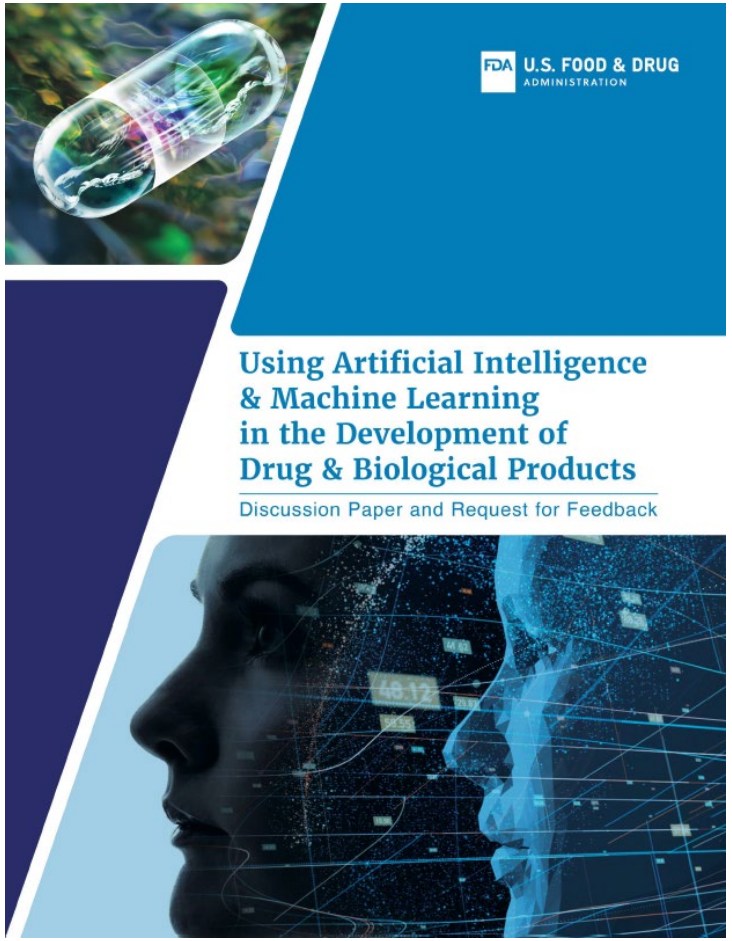
ML has been applied to various areas in drug safety evaluation using real world data - from pattern identification to causal inference.



Zhang et al., 2022. The Use of Machine Learning in Regulatory Drug Safety Evaluation. Statistics in Biopharmaceutical Research, pp.1-13.

FDA's perspective on the use of AI/ML in drug development

- FDA recognizes the increased use of AI/ML throughout the drug development life cycle and across a range of therapeutic areas.
- FDA CDER, CBER and CDRH issued a **discussion paper** to explore relevant considerations for the use of AI/ML in the development of drugs and biological products.
- This paper describes four areas of focus for CDER, CBER and CDRH, and OCP regarding the development and use of AI across the medical product life cycle.



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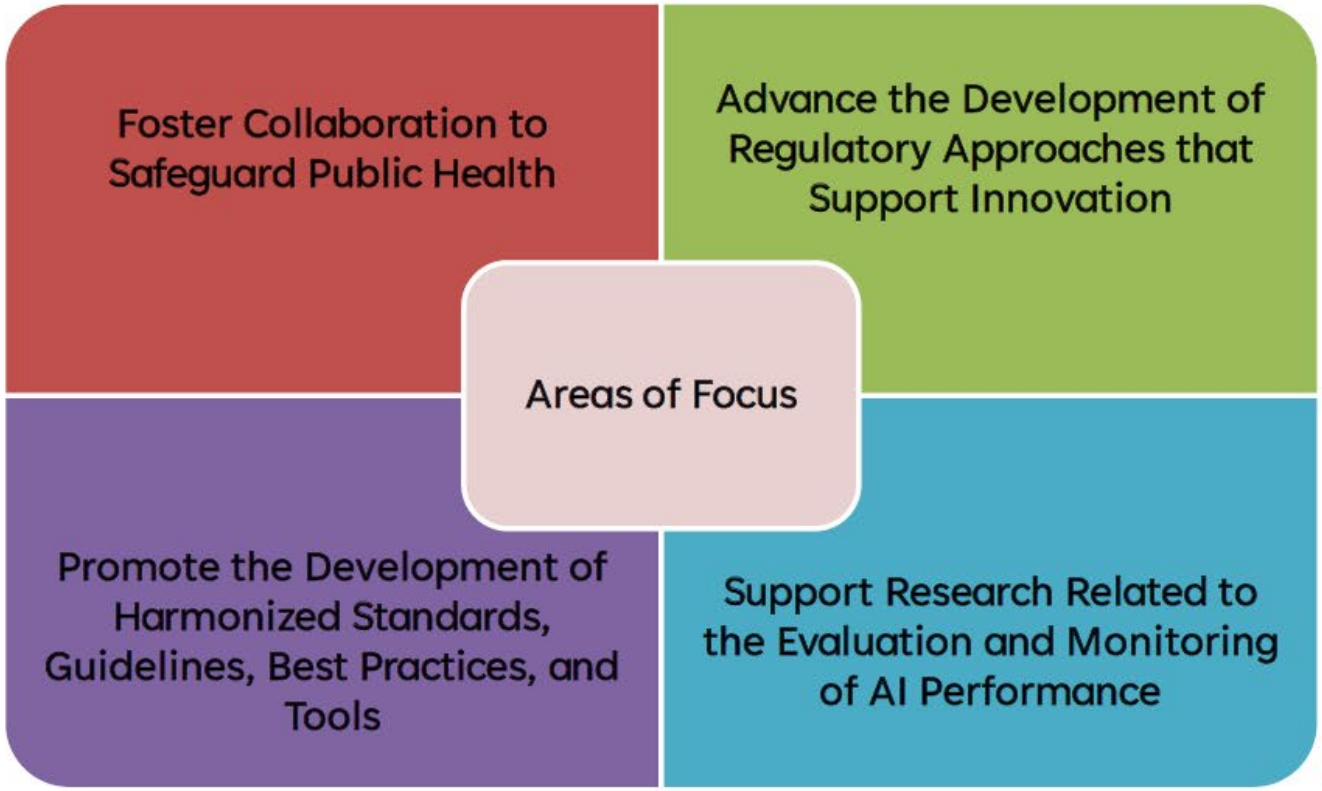

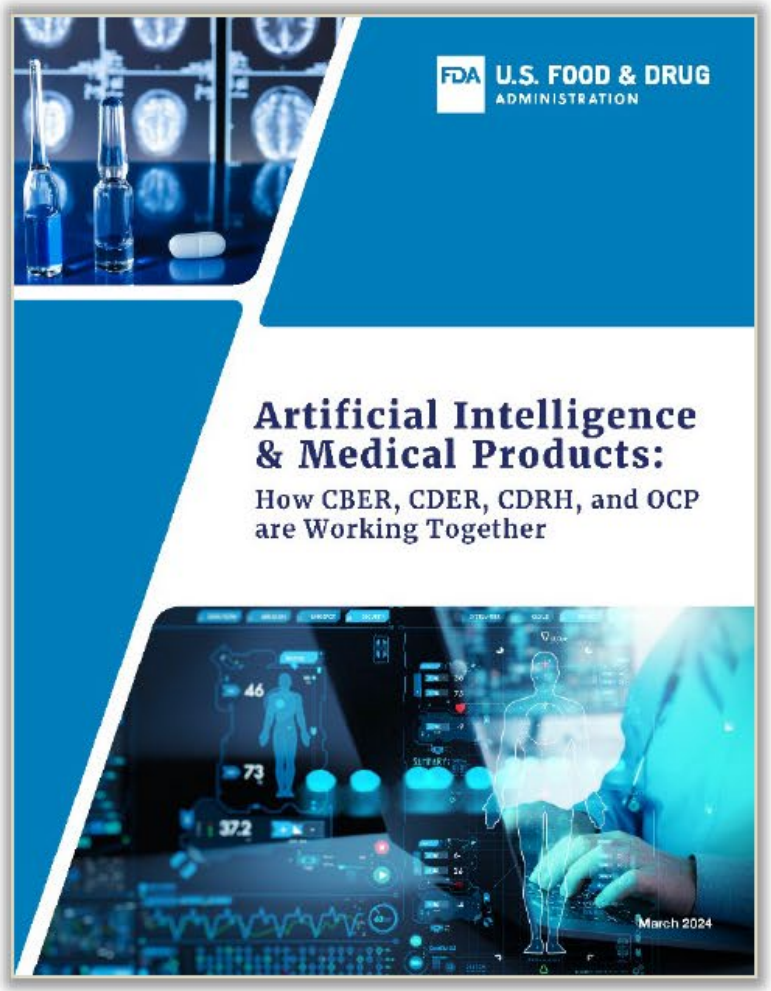

https://www.fda.gov/media/177030/download?attachment

Part 3

Challenges and Efforts

Challenges

- **Data, data, data:**
  - Scarcity of high-quality, large-scale, and fit-for-purpose datasets
  - Identification and mitigation of bias in datasets
  - poor generalization due to dataset shift
- **Interpretability and explainability**
  - Black box algorithms
  - Need for regulatory clarity in certain areas
  - When these two are the same and different? When these matter and when not?
- **Transparency and governance**
  - Ensuring transparency and reproducibility to regulators and other potential users
  - Providing oversight/governance for adaptive algorithm

FDA/CDER Efforts

- CDER actively collaborates with CDRH, CBER, and other centers on issues related to AI/ML in medical product development to ensure consistency whenever possible
- CDER established an AI Steering Committee in 2020 to facilitate effective use and sustainment of AI in CDER's decision-making and operations
- Next steps: workshops, white papers, discussion paper, harmonizing and tailoring CDRH's guidance on good ML practice to drug development, support development of regulatory science methods

- **Goal: to develop and adopt a flexible risk-based regulatory framework that promotes innovation and protects patient safety**

What's next?

- Continue responsive risk-based regulation approach
  - An iterative approach that keeps the pace with this rapidly evolving technology
- Continue engaging all parties across the AI ecosystem (i.e., academia, industry, biotech, international regulators, etc.)
- Advance regulatory science in this area

Thank you! Questions and Comments?

Hana.Lee@fda.hhs.gov

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